

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. CHB-602: Inorganic Chemistry-IV

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following:

[5 × 2 = 10]

- (a) Why are low-spin complexes of Ni(II) diamagnetic (0.0 B.M.) whereas high-spin complexes are paramagnetic (2.9 B.M.)?
- (b) Draw the structure of the compound obtained by the reaction of uranyl nitrate with sodium acetate in acetic acid.
- (c) Compare the electronic configurations of the first five actinides with those of the corresponding lanthanides.
- (d) Describe the nature of the Co-C bond present in methylcobalamin (Vitamin B₁₂).
- (e) What is the role of the sodium-potassium pump in biological systems?

Question 2.

- (a) Why do Pu(III) and Am(III) show anomalous magnetic properties? Explain in terms of spin-orbit coupling. [6]
- (b) Explain the observed molar extinction coefficients (ϵ in L mol⁻¹ cm⁻¹) for the following complexes: [8]

[TiCl₆]²⁻ (10000), [CoCl₄]²⁻ (500), [Ti(H₂O)₆]³⁺ (10), [MnBr₄]²⁻ (4), [Mn(H₂O)₆]²⁺ (0.02)

Question 3.

- (a) Describe the Gouy method for the determination of magnetic susceptibility of a complex. Compare it with the Faraday method. [8]
- (b) Account for the following: [6]
 - (i) KMnO₄ shows an intense violet color despite having a d⁰ metal center.
 - (ii) HgCl₂ is white while HgI₂ has a dark red color.

Question 4.

- (a) What are the selection rules for d-d electronic transitions? Describe the mechanisms of relaxation of these rules. [6]
- (b) Give a brief account of Charge Transfer (CT) transitions in the electronic spectra of coordination compounds. Differentiate between LMCT and MLCT with examples. [8]
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Question 5.

- (a) How is thorium extracted from monazite sand ore? Write down all reactions. [6]
- (b) How are transuranic elements separated from the spent fuel rods of a nuclear reactor (PUREX process)? [8]